

Product Recalls and Brand Architecture

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The illusion of choice



Brand architecture

- Firms choose how to name their products

Umbrella / branded house

one name across products



Kellogg's

Individual / house of brands

separate names, parent hidden



PepsiCo (Frito-Lay)

- Shared names transfer the good
 - Toothpaste ads lift same-brand toothbrush sales (Erdem & Sun 2002)
- But they may also transfer the bad
 - Audi 5000 acceleration scare cut resale prices of other Audi models (Sullivan 1990)

Product recalls

- A recall is a discrete, public, quality-related shock to one product
- Does it spill over to the rest of the brand?
 - same brand, same category
 - same brand, other categories
 - sister brands owned by the same parent
- Spillovers may be governed by ownership, shared production, category, etc.
 - If shared name is important, brand architecture should moderate them



Literature: Recall-ish events and within-firm spillovers

Paper	Shock	Spills to	Sign
Freedman et al. (<i>REStat</i> 2012)	Toy recall	Same firm's other toys	None
Liu & Shankar (<i>Mgmt Sci</i> 2015)	Car recall	Sub-brands, same name	Negative
Borah & Tellis (<i>JMR</i> 2016)	Car recall	Other models, same name	Negative
Giannetti & Srinivasan (<i>ML</i> 2021)	Car recall	Same/sister brand	Mixed
Che et al. (<i>JMR</i> 2023)	VW scandal	Compliant VW models	Negative
He (<i>ERAE</i> 2025)	Fish recall	Other brands, same species	Negative
Keller & van Heerde (<i>JM</i> 2026)	Deletion	Sister brands, other names	Positive

- Contradictory signs across single-event studies, no variation in brand architecture
- Freedman et al.: “If consumers do not know that toys produced under the Fisher-Price brand are, in fact, produced by Mattel, they will not be able to accurately act upon that inference.”

Research questions

1. Does negative information spill over across a brand's portfolio?
2. Does brand architecture moderate the impact of spillovers?

Empirical context: food recalls

- Many events: $\sim 2,000$ FDA enforcement records per year, plus USDA FSIS
- Packaged food: brands and parents matter
- Variation in architecture across CPG parents, and within parent

Data

- **Recalls:** FDA and FSIS, 2012-
- **Sales:** NielsenIQ Retail Scanner, UPC $\times$ store $\times$ week
- **Recall $\rightarrow$ UPC linkage**
 - barcodes in notice text, press pages, description lookup; tiered by confidence
 - master file: 13,783 recall-UPC rows; $\sim 35\%$ of FDA events at high confidence why
- **Brand $\rightarrow$ parent map:** hand-built (not in Nielsen) [in progress]
- **Architecture:** front-of-pack endorsement coding [in progress]

Spillover rings

Ring	Relation to recalled product	Category
0	Recalled UPC	Recalled
1a	Same brand	Recalled
1b	Same brand	Other
2a	Sister brand, same parent	Recalled
2b	Sister brand, same parent	Other
3a	Outsider	Recalled
3b	Outsider	Other

- Spillovers vary with proximity to recalled product in corporate distance $\times$ category distance space

Estimator

Simple version: do the sales of affected products change after the recall, relative to comparison products (difference-in-differences)?

$$q_{jt} = \alpha_j + \lambda_t + \beta(D_j \times \text{Post}_t) + \varepsilon_{jt}.$$

- j : UPC, t : week, D_j : affected UPC, Post_t : post-recall

Fancy version: Stacked event study with discrete outcome (PPML)

$$\mathbb{E}[q_{jset}] = \exp \left(\alpha_{jse} + \lambda_{c(j)tse} + \sum_r \sum_{k \neq -1} \beta_{rk} D_{je}^r \mathbf{1}[t - t_e = k] \right)$$

- $c(j)$: category, s : store, e : recall event, t_e : event time, D_{je}^r : ring indicator
- β_{rk} is identified by changes in the sales gap between affected and comparison products within the same store and category, around the same recall event

Estimator

Fancy version with brand architecture moderator:

$$\mathbb{E}[q_{jset}] = \exp \left(\alpha_{jse} + \lambda_{c(j)tse} + \sum_r \sum_{k \neq -1} (\beta_{rk} + \gamma_{rk} A_{je}) D_{je}^r \mathbf{1}\{t - t_e = k\} \right).$$

- A_{je} : Does UPC j display the parent brand's name on its packaging?

Identification

- β_{rk} is identified by changes in the sales gap between affected and comparison products within the same store and category, around the same recall event
- Causal interpretation requires that, absent the recall, this sales gap would have remained stable, conditional on FEs
- Considerations
 - Anticipation (timing)
 - Effect on comparison products (relative vs. absolute difference)
 - Correlated shocks (e.g., production problems that would reduce sales even without the recall)
 - Retailer vs. consumer behavior

Initial evidence

- I thought I might have something ready to present here, but alas...

Appendix: why only ~35% of FDA events link to a UPC

FDA events, 2012–2024 ($n = 6,989$)	Share
Linked, high confidence	36%
national brand	30%
store brand only (hidden in Nielsen store data)	6%
Linked, low confidence only	3%
Not in Nielsen	~58%
barcode in notice, not a Nielsen product	14%
not retail: foodservice, bulk, ingredient	~15%
small / regional / import / direct-to-consumer	~13%
store brand, no barcode in notice	~8%
fresh, deli, random weight	~4%
dietary supplements	~2%
notice states no UPC	1%
National brand, likely in Nielsen, unlinked	~3%

- Most misses are products Nielsen does not carry, not failed matches
- Among recalls of products Nielsen carries: roughly 36 of ~39
- The most serious recalls (FDA Class I) split the same way
- All FDA events (7,817) include 828 outside 2012–2024 $\Rightarrow$ 35% headline